PARALLEL LINES

8–1 DEFINITION OF PARALLEL LINES

In earlier mathematics courses, you learned some facts about parallel lines. Before reading this chapter, try to recall some properties of parallel lines that you have accepted as true. The following exercises will help your memory and prepare you for proofs.

Exercise Group 8–1

1. What do you mean when you say, "Line *l* is *parallel* to line *m*"?
2. Name some geometric figures that have pairs of *parallel* sides.
3. Explain why two sides of a triangle are not *parallel*.
4. Two rocks are dropped, one from the Brooklyn Bridge, and the other from the Golden Gate Bridge. Would you say that these rocks fall along *parallel* paths?
5. If you drop two rocks, one from each hand, do they traverse *parallel* paths?

6. Roads r and s run "due north and south." Are they *parallel*?
7. Roads t and u run "due east and west." Are they *parallel*?
8. How many different pairs of *parallel* edges does a cube have?
9. Both lines l and $m \perp$ line n. Does this suggest a theorem about *parallels*?
10. Each of the lines l and m is *parallel* to line n. What theorem does this suggest?
11. Lines p and q are *parallel*. Line $r \perp p$. What theorem does this suggest?
12. Draw two angles, $\angle AOB$ and $\angle A'O'B'$, such that the lines AO and $A'O'$ are *parallel*, and also lines BO and $B'O'$ are *parallel*. Does this figure suggest a theorem? Be careful! Draw more than one figure.
13. Draw a line l, and mark a point A not on l. What might it mean to say that a ray from A is *parallel* to line l?
14. Which of the above exercises do not strictly belong to plane geometry?

If you discussed Exercises 6 and 7, someone probably pointed out that roads on the surface of the earth are not quite the same as lines. Also, in connection with Exercises 4 and 5, the paths along which objects fall are hardly the imaginary lines of our geometry. Since the earth rotates about its axis and at the same time revolves around the sun, an object falling toward the center of the earth follows a path extremely hard to visualize. Our task is to choose a definition of parallel lines, such that we can use the definition in our plane geometry along with our postulates to prove the truth of statements that concern parallel lines and that we feel ought to be true. The definition below was perhaps suggested by someone in the class during the discussion of Exercise Group 8–1.

Definition 8–1. Two lines l and m are parallel if and only if they do not intersect. We indicate that two lines are parallel by writing "$l \parallel m$."

Exercise Group 8–2

1. Give definitions of the following. In later sections, it will be convenient to have these definitions. Good definitions are not hard to make, and so are given as exercises here.
   (a) A ray parallel to a line.
   (b) Parallel rays.
   (c) A line segment parallel to a line.
   (d) Parallel line segments.
2. Draw a figure of a line not intersecting a segment and not parallel to the segment.
3. Can a ray and a line not intersect and yet not be parallel?

Now that we have parallel lines, it is possible for us to talk about rays that have the same direction. This requires another definition.

Definition 8–2. Two rays have the same *direction* if and only if they are parallel and both rays are on the same side of the line through their vertices (Fig. 8–1).

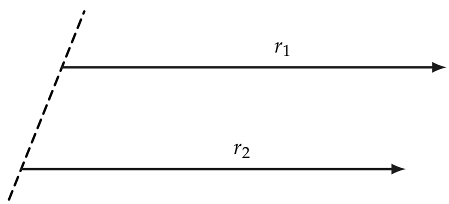

FIGURE 8–1

Exercise 8–3

What would have been wrong had we defined lines to be parallel if they had the same direction?

8–2 EXISTENCE OF PARALLELS

Perhaps it will seem strange to you that we are going to prove that parallel lines actually exist. We could add an extra postulate and assume that there are parallels, but the proof is not difficult. We give three different proofs of the following theorem.

Theorem 8–1. *If l is any line, and A any point not on l, then there exists at least one line through A parallel to l.*

First Proof. Referring to Fig. 8–2, let B be any point on l, and let n be the line through A and B. Let E be another point on l, and let D be a point on the same side of n as E, such that $\angle DAC \cong \angle EBA$. Lines l and m cannot intersect, for if they did, say at P, we should have $\triangle PBA$ with exterior $\angle PAC \cong \angle PBA$. But we know this is impossible, for an exterior angle of a triangle is always greater than either opposite interior angle.

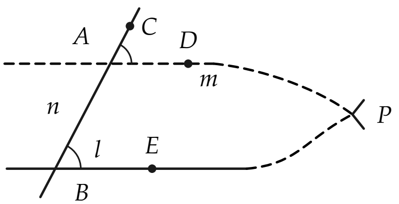

FIGURE 8–2

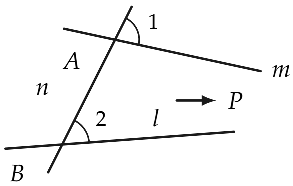

FIGURE 8–3

**Second Proof. Lemma.* Let line l, point A, and line n be as in Theorem 8–1. Now, referring to Fig. 8–2, if m is not parallel to

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* If the teacher wishes, the second and third proofs may be omitted. Note that the second proof employs a lemma, or helping theorem.

l, then $\angle 1$ is not congruent to $\angle 2$. (For "not parallel to," we use the symbol $\nparallel$.) *Proof.* If m is not parallel to l, these lines intersect, a triangle is formed, and $\angle 1 \ncong \angle 2$. Now Theorem 8–1 follows quickly from the lemma. The lemma states that $l \nparallel m \to \angle 1 \ncong \angle 2$. The contrapositive of the lemma is $\angle 1 \cong \angle 2 \to l \parallel m$, and the theorem obviously follows.

**Third Proof.* Referring to Fig. 8–4, let l, m, and n refer to the lines of the preceding proofs with n cutting m at A, and l at B, and the indicated angles at A and B congruent to each other. If l and m intersected at P, we could choose Q on m, as above, so that $AQ \cong BP$. It follows at once, by the S.A.S. theorem, that $\triangle BAQ \cong \triangle ABP$. Hence, $\angle ABQ \cong \angle BAP$.

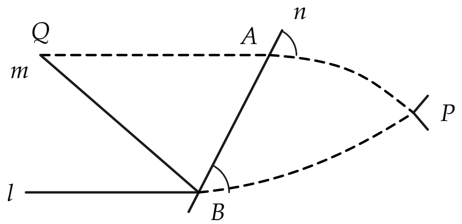

FIGURE 8–4

Since also, $\angle ABP \cong \angle BAQ$, we have $\angle QBP \cong \angle QAP$. $\angle QAP$ is a straight angle, so $\angle QBP$ is a straight angle. It follows that Q lies on line l, and so if lines l and m intersect in one point, they have a second point in common. This is impossible, so lines l and m do not intersect.

Exercise Group 8–4

1. Draw a line, and mark a point not on it. Use the first proof of Theorem 8–1 to draw a line, parallel to the first line, through your point.

2. Prove that if $l \perp m$, and $n \perp m$, then, if $l \neq n$, $l \parallel n$.
3. Is a line parallel to itself?
4. In the first proof of Theorem 8–1, several details were omitted. First a point B was chosen and then a point E. How was C selected? How would we have argued if l and m were to intersect on the other side of AB?
5. The second proof of Theorem 8–1 is really just a reworded version of the first proof. Explain why this is so.
6. Without pointing at Fig. 8–4, describe how the point Q is chosen in the third proof. What postulate guarantees the existence of Q?
7. Prove that if $l_1 \parallel l_2$, then all the points of l_2 are on the same side of l_1.
8. Use the result of Exercise 7 to give a definition of "the set of points between l_1 and l_2, where $l_1 \parallel l_2$."

8–3 THE PARALLEL POSTULATE

We proved in Section 8–2 that through a point A not on line l, there is *at least one line parallel to* l. Now we raise the question, *is there more than one such line?* Our geometric intuition tells us that there is only the one, and it is natural to attempt to prove this from our other postulates. For 2000 years, many fine mathematicians tried and failed to find such a proof. At last, in the 19th century, it was shown that *it is impossible to prove from the postulates we have already listed that there is no more than one line through a given point parallel to a given line*. Euclid himself assumed that there was no more than one such parallel, and we shall make the same assumption.

Postulate VI–1. *The parallel postulate.* Through a point not on a line l, there is no more than one line parallel to l.

Historical note. It may seem strange that it is possible to prove that the parallel postulate is not a consequence of the other postulates. The understanding of the fact that the parallel postulate is independent of the others came about in the following way.

After vainly trying to prove the truth of the parallel postulate by using the other postulates of Euclid, Johann Bolyai (Yo′-hahn Bull′-yī), a young Hungarian mathematician, decided to assume that *a second parallel could be drawn through A*. Using this postulate, he proved many strange and beautiful theorems. Later it was shown that Bolyai's geometry was just as acceptable as ordinary Euclidean geometry. We call this geometry of Bolyai's a *non-Euclidean* geometry to contrast it with our Euclidean geometry which employs the parallel postulate of Euclid. Whenever you encounter the term non-Euclidean geometry, you can be sure that some postulate different from the parallel postulate is used. All the theorems we have proved up to now are true in the non-Euclidean geometry of Bolyai.

When these things are first explained, many students ask which postulate is right. Can only one parallel be drawn, or is it possible to draw several? While these questions are quite natural ones to ask, they do not have the simple answers that one might expect. Our postulates are statements that we invent in order to prove interesting and useful theorems. The theorems of Euclidean geometry are useful in carpentry, drafting, surveying, navigation, astronomy, and in many other fields. The theorems of some non-Euclidean geometries have been found to be extremely useful in astronomy and physics. Today mathematicians study many different geometries. Each of the geometries has a postulate system different from that of any other geometry. All are interesting. Many have proved to be of practical use. It does not make sense to ask which geometry is true. Mathematicians design geometries as architects design houses. You might as well ask which house is true. It *is* appropriate to ask which house is better adapted to one's living requirements. And so it is also appropriate to ask which geometry is most convenient to describe the universe.*

KARL FRIEDRICH GAUSS
(1777–1855)
Probably the first mathematician to recognize the existence of non-Euclidean geometries.

Courtesy of *Scripta Mathematica*.

FIGURE — (book p. 137)

*See, for example, the discussion of Gauss' experiment in *The World of Mathematics*, by J. R. Newman, Vol. 3, p. 1644.

NICOLAI IVANOVICH LOBACHEVSKY
(1793–1856)
Co-inventor, with Johann Bolyai, of
non-Euclidean geometry.
Courtesy of *Scripta Mathematica*.

FIGURE — (book p. 138)

8–4 CONSEQUENCES OF THE PARALLEL POSTULATE

We can now prove several interesting theorems quite easily.

Theorem 8–2. *If l, m, and n are distinct lines, with $l \parallel m$ and $m \parallel n$, then $l \parallel n$* (Fig. 8–5).

Proof. If l were not parallel to n, then l and n would intersect, say at P in Fig. 8–5. Two lines, namely l and n, would then pass through P, parallel to m, thus contradicting the parallel postulate.

Theorem 8–3. *If $l \parallel m$, and line $n \neq l$ intersects l at P, then n intersects m* (Fig. 8–6).

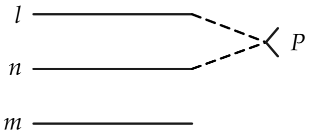

FIGURE 8–5

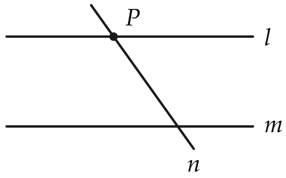

FIGURE 8–6

Proof.

STATEMENTS	REASONS
1. $l \parallel m$.	Given.
2. P is on n and l, and $n \neq l$.	Given.
3. l is the only line on P parallel to m.	Statements 1, 2, and the parallel postulate.
4. n intersects m.	Statements 2 and 3.

Exercise Group 8–5

1. Prove that the construction of a parallel to l through a given point A is an RC construction.
2. In Fig. 8–7, $\angle ABC \cong \angle BCD$. Prove that $AB \parallel CD$.

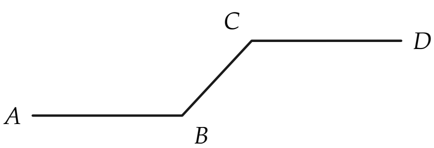

FIGURE 8–7

3. Prove that if $l \perp n$, $n \perp m$, and $l \neq m$, then $l \parallel m$. Have you used the parallel postulate?
4. If $l \parallel m$, and $l \perp n$, prove that $n \perp m$. Did you use the parallel postulate?
5. If $l \parallel m$, $k \parallel m$, and l and k intersect at P, what can you say?

*6. In Fig. 8–8, $\overline{AB} = \overline{CD}$, and $\angle ABC$ and $\angle BCD$ are right. Prove that $AD \parallel BC$. [*Hint:* Prove that $\angle A \cong \angle D$.]

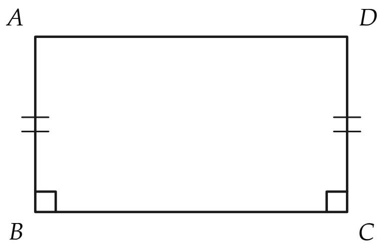

FIGURE 8–8

*7. If a rectangle is defined as a quadrilateral with four right angles, prove that rectangles actually exist.

8–5 CORRESPONDING AND ALTERNATE ANGLES

When a line intersects two other lines, we often wish to refer to certain pairs of the angles formed. In order to be precise, we need a definition.

Definition 8–3. A line l (Fig. 8–9) that intersects lines l_1 and l_2 in distinct points is called a *transversal* of l_1 and l_2. We also say that l_1 and l_2 are cut by the transversal l. Angles 3, 4, 5, and 6 are called *interior angles*. Angles 1, 2, 7, and 8 are called *exterior angles*. These eight angles are grouped into four pairs referred to as pairs of *corresponding angles*. Angles 1 and 5 form such a pair of corresponding angles, as do also 2 and 6, 3 and 7, and 4 and 8. Angles such as 3 and 5 are called *alternate angles*. Other pairs of alternate angles are 2 and 8, 1 and 7, and 4 and 6. We speak of the pair 3 and 5 as *alternate interior angles*. We call the pair 2 and 8 *alternate exterior angles*. We observe that *alternate* angles have the points of their interiors on *opposite* sides of l, while *corresponding* angles have their interiors on the *same* side of l. In a pair of corresponding angles, one angle is always exterior and one interior.

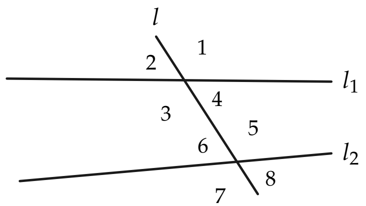

FIGURE 8–9

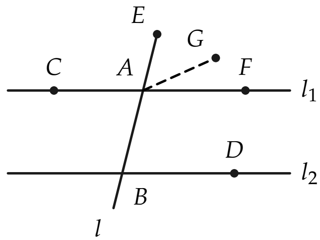

FIGURE 8–10

Theorem 8–4. *Let l_1 and l_2 be lines cut by line l. If $l_1 \parallel l_2$, then two alternate interior angles are congruent to each other, and conversely, if two alternate interior angles are congruent, then $l_1 \parallel l_2$* (Fig. 8–10).

Proof. The last part of the theorem follows at once from the first proof of Theorem 8–1, for if $\angle CAB \cong \angle ABD$, then $\angle ABD \cong \angle EAF$, and $l_1 \parallel l_2$.

The first part of the theorem requires the parallel postulate, and may be proved as follows. Choose G, located in the interior of $\angle DBE$, such that $\angle EAG \cong \angle ABD$. Then line $AG \parallel l_2$. (Why?) But A is also on l_1, and $l_1 \parallel l_2$. Hence the line through A and G is the same as l_1, so G is on l_1. Now, $\angle CAB \cong \angle EAG \cong \angle ABD$. Thus the alternate interior angles are congruent.

As important consequences of Theorem 8–4, we state the following corollaries, which you can easily prove.

Corollary 8–4–1. Let l cut l_1 and l_2 at distinct points. If $l_1 \parallel l_2$, then alternate exterior angles are congruent, and conversely.

Corollary 8–4–2. Let l cut l_1 and l_2 at distinct points. If $l_1 \parallel l_2$, then corresponding angles are congruent, and conversely.

Exercise Group 8–6

1. Prove that if a line is perpendicular to one of two parallel lines, it is perpendicular to the other.
2. Prove that if $l \parallel m$, $n \perp l$, $p \perp m$, and $n \neq p$, then $n \parallel p$.
3. In Fig. 8–11, $\overline{AO} = \overline{CO}$, and $\overline{BO} = \overline{OD}$. Prove that $AB \parallel CD$.

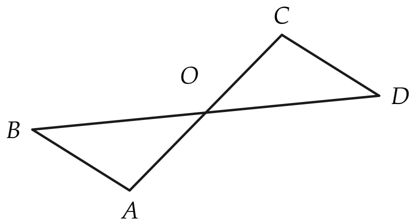

FIGURE 8–11

4. Show that $\overline{BC} = \overline{AD}$, and $BC \parallel AD$ in Fig. 8–11.
5. Show that if $AB \cong CD$, and $AB \parallel CD$, then $BO \cong OD$, and $BC \parallel AD$ in Fig. 8–11.
6. If $l \parallel m$, and A and B are any two points on l, show that the perpendicular distances from A and B to m are equal. We may state this result in the form: if $l \parallel m$, then l and m are "everywhere equidistant." State and prove the converse of this theorem. State also the contrapositive.

8–6 ANGLE SUMS

One of the most interesting consequences of the parallel postulate is a theorem relating to the angles of a triangle. We need a definition.

Definition 8–4. If $\angle AOB$ and $\angle BOC$ are adjacent angles (Fig. 8–12), and either $\angle AOC$ is a straight angle, or OB is in the interior of $\angle AOC$, then we say that $\angle AOC$ is a *sum* of $\angle AOB$ and $\angle BOC$. Likewise, if $\angle A'O'B' \cong \angle AOB$, and $\angle B''O''C'' \cong \angle BOC$, then we say that $\angle AOC$ is a *sum* of $\angle A'O'B'$ and $\angle B''O''C''$.

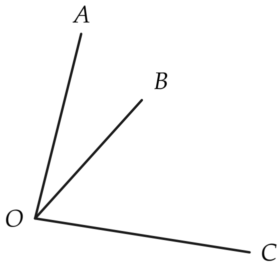

FIGURE 8–12

Remark. At this stage in our geometry, we cannot yet "add" angles by adding their measures in degrees, because we have not as yet described how angles are measured. The measurement of angles is more complicated than measurement of length, and will be described in a later chapter.

Draw a figure to illustrate the last part of Definition 8–4.

Definition 8–5. Two angles are said to be *complementary* if their sum is a right angle.

Definition 8–6. Two angles are said to be *supplementary* if their sum is a straight angle.

Exercise Group 8–7

1. Draw an obtuse angle. Draw an angle that is supplementary to this angle and is not adjacent to it. Draw one that is adjacent and supplementary.
2. Draw an acute angle. Draw an angle that is complementary to this angle and is adjacent to it. Draw one that is not adjacent, but is complementary, to the acute angle.
3. Prove that if two acute angles are congruent, then their complements are congruent.
4. Prove that if the sides of one angle are parallel, respectively, to the sides of a second angle, then either the angles are congruent or they are supplementary.

Theorem 8–5. *The sum of the angles of a triangle is a straight angle.*

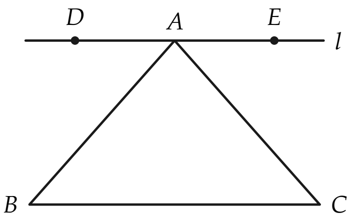

FIGURE 8–13

Proof. In Fig. 8–13, let l be the line through A and parallel to the side BC of $\triangle ABC$. Let D be a point on l that is on the opposite side of AB from C. Let E be a point of l on the same side of AB as C. Then $\angle DAB \cong \angle ABC$, and $\angle EAC \cong \angle ACB$. The straight angle DAE is the sum of angles A, B, and C.

Exercise Group 8–8

1. Show that the acute angles of a right triangle are complementary.

2. Two angles of a triangle are $40^\circ\ 17'\ 30''$ and $78^\circ\ 31'\ 38''$, respectively. Determine the number of degrees in the third angle.

3. How many degrees are there in each angle of an equilateral triangle?

4. Show that a median of an equilateral triangle divides the triangle into two right triangles, each of which has acute angles of 30° and 60°.

5. Prove that in a right triangle, with acute angles of 30° and 60°, the side opposite the 30° angle is half as long as the hypotenuse.

6. In $\triangle ABC$, an exterior angle at B is 124°. $\angle A$ is 58°. How many degrees in $\angle C$?

7. Prove that if two angles of one triangle are congruent, respectively, to two angles of another triangle, then the third angles are congruent.

8. In Fig. 8–14, $AB \cong A'B'$, $\angle A \cong \angle A'$, and $\angle C \cong \angle C'$. Prove

that $\triangle ABC \cong \triangle A'B'C'$.

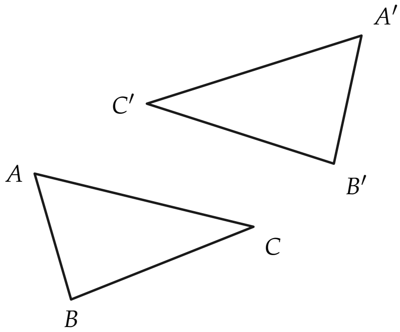

FIGURE 8–14

9. Is the following true? Two triangles are congruent to each other if one side and two angles of the one triangle are congruent to one side and two angles of the other.

10. Prove that two right triangles are congruent if the hypotenuse and an acute angle of one triangle are congruent, respectively, to the hypotenuse and acute angle of the other.

11. In $\triangle ABC$ (Fig. 8–15), $\angle A$ is right, and $AD \perp BC$. Prove that $\angle BAD \cong \angle ACD$, and $\angle ABD \cong \angle DAC$.

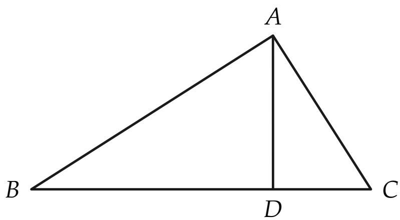

FIGURE 8–15

*12. In Fig. 8–16, D is the midpoint of BC, and $BD \cong AD$. Prove that $\angle BAC$ is right.

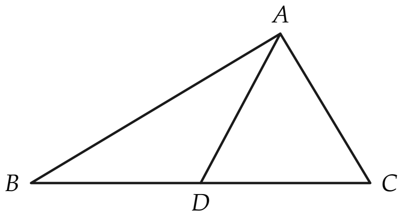

FIGURE 8–16

*13. Prove that two right triangles are congruent if the hypotenuse and one side of one triangle are congruent to the hypotenuse and one side of the other.

8–7 POLYGONS

In this and the remaining sections, we will be studying certain polygons. First, we need to recall the definition of a polygonal path.

A *polygonal path* (Fig. 8–17) is determined by a number of points, $P_1, P_2, \ldots, P_n$, called the *vertices*, given in a definite order. The path is the set of points on the segments $P_1P_2, P_2P_3, \ldots, P_{n-1}P_n$. The point P_1 is called the *beginning* of the path, and P_n is called the *end*. The segments P_1P_2, etc., are called the *sides* of the path.

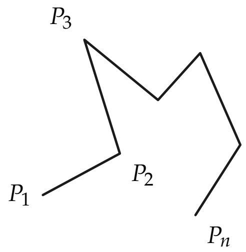

FIGURE 8–17

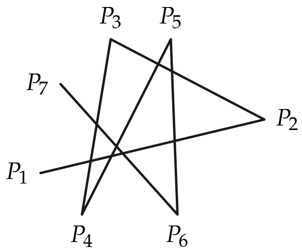

FIGURE 8–18

Remark. There is no requirement that the sides of a polygonal path not intersect. Hence the path can be quite complicated. See Fig. 8–18 for a six-sided path.

Exercise Group 8–9

1. Draw a polygonal path with five sides that does not cross itself. Draw one that does.
2. If the length of a polygonal path is the sum of the lengths of its sides, how does the length of a polygonal path $P_1, \ldots, P_n$ compare with $\overline{P_1 P_n}$?

Definition 8–7. A *polygon* is a polygonal path whose beginning and end coincide. If the vertices are all different and no two sides have a point in common (other than a vertex of two adjacent sides), then the polygon is called *simple* (Fig. 8–19).

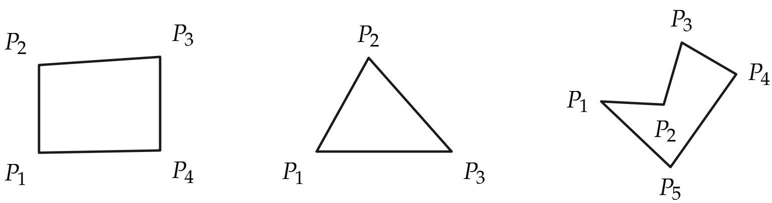

FIGURE 8–19. Simple polygons.

Each simple polygon separates the points of the plane, other than the polygon itself, into two sets called the *interior* and *exterior* of the polygon. If A is a point of the interior, and B is a point of the exterior, every polygonal path connecting A to B contains at least one point of the polygon. If A and B both belong to either the interior or exterior, then there exist polygonal paths joining A to B that contain no point of the polygon. The proofs of these statements are quite difficult, and will not be given here. It would be instructive to carry out the proofs in the case in which the polygon is a triangle. The student interested in doing so should restudy Section 3–6.

**Problem.* The theorem that a simple polygon separates the plane is powerful and has surprising applications. It can be used to show that the well-known puzzle below has no solution.

It is desired to connect each of three houses A, B, and C with three outlets, one for gas G, one for water W, and one for electricity E.

$$\begin{array}{ccc} A & B & C \\ \cdot & \cdot & \cdot \\ G & W & E \\ \cdot & \cdot & \cdot \end{array}$$

Is it possible to run the lines, one from each house to each outlet, in such a way that no lines cross? In other words, can polygonal paths from A to G, A to W, etc., be drawn so that no two paths cross?

Definition 8–8. Vertices of a polygon that are endpoints of the same side are called *adjacent* vertices (Fig. 8–20). Angles whose vertices are adjacent to each other are called *consecutive* angles. A *diagonal* of a polygon is a segment joining any two nonadjacent vertices. Sides with a common vertex are called *adjacent* sides.

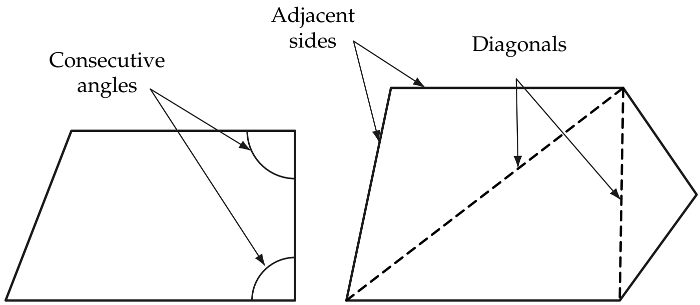

FIGURE 8–20

Definition 8–9. A simple polygon is called *convex* if, whenever A and B are points of the polygon, every point of the segment AB lies either on the polygon or in its interior (see Fig. 8–21).

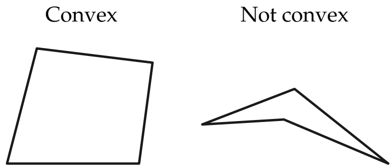

FIGURE 8–21

Remark. Unless otherwise stated, we will confine our attention to simple convex polygons.

Definition 8–10. If all sides of a polygon have the same length, the polygon is *equilateral*. If all angles are congruent, it is *equiangular*. A convex polygon that is equilateral and equiangular is called *regular*. Polygons are also classified according to the number of sides. Polygons of 3, 4, 5, 6, 7, 8, 9, and 10 sides are called *triangles*, *quadrilaterals*, *pentagons*, *hexagons*, *heptagons*, *octagons*, *nonagons*, and *decagons*, respectively. A polygon of n sides will be called an *n-gon*.

Exercise Group 8–10

1. What do we call a regular polygon of three sides? of four sides?
2. Draw a convex polygon of five sides. Draw a quadrilateral that is not convex. Draw a polygon that is not simple.

*3. Is it true that every interior point of a convex polygon is also interior to every angle of the polygon?

4. We can divide a convex polygon into triangles by drawing the diagonals from any vertex. Into how many triangles does this procedure divide a convex polygon of four sides? of ten sides? of n sides?

*5. How many diagonals has a convex polygon of four sides? five sides? ten sides? n sides?

6. Show that an equiangular polygon need not be equilateral.

*7. Find an example of an equilateral, equiangular polygon that is not regular.

A diagonal of a convex quadrilateral divides the quadrilateral into two triangles. Since the sum of the angles in each triangle is a straight angle, we shall say that *the sum of the (four) angles of the quadrilateral is two straight angles*. The two diagonals from one vertex of a convex pentagon divide the pentagon into three triangles, and we say that *the sum of the (five) angles of the pentagon is three straight angles*.

When we have defined degree measure for angles, we will see that the sum of the degree measures of the angles of a convex quadrilateral is $2 \cdot 180^\circ = 360^\circ$, of a pentagon, $3 \cdot 180^\circ = 540^\circ$, and in general for a polygon of n sides, the sum of degree measures is $(n-2) \cdot 180^\circ$.

Do not misunderstand the statement that the sum of the angles of a pentagon is three straight angles. This is using the word "sum" in a different sense than when we speak of the sum of *two angles*. Recall that when we add two angles, the sum is less than or congruent to a straight angle.

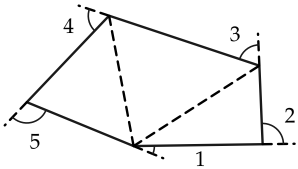

FIGURE 8–22. Pentagon (three triangles).

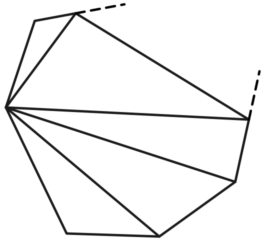

FIGURE 8–23. n-gon ($n-2$ triangles).

We observe that the sum of the five interior angles and the five indicated exterior angles in Fig. 8–22 is precisely five straight angles. Since the interior angles account for three straight angles, the sum of the exterior angles must be two straight angles. Similar reasoning shows that for any convex polygon of n sides, the sum of the n exterior angles (one at each vertex) is exactly two straight angles (Fig. 8–23).

Exercise Group 8–1

1. A quadrilateral has three 95° angles. How many degrees are there in the fourth angle?
2. How many degrees are there in each angle of a regular hexagon?
3. How many degrees are there in each exterior angle of a regular pentagon?
4. A convex quadrilateral has three angles of the same size. The fourth angle is a 120° angle. How many degrees are there in each of the other angles?

5. Explain why it is impossible for a convex polygon to have as many as six angles smaller than 120°.

*6. What is the sum of the angles at the vertices of a regular five-pointed star? (The points of the star are vertices of a regular pentagon.)

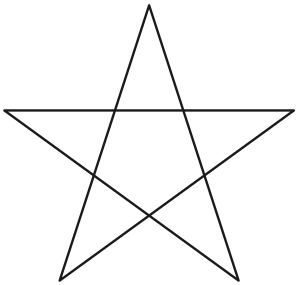

FIGURE 8–24

7. An exterior angle of a regular polygon is 20°. How many sides has the polygon?

8. Pentagon $ABCDE$ is regular. Prove that a line l, bisecting the angle at A, is perpendicular to side CD.

9. How many sides does a regular polygon have if each angle is 120°? 140°? 160°? 179°?

10. Make up some theorems of your own relating to regular pentagons or regular hexagons.

8–8 PARALLELOGRAMS

The following definitions make certain familiar terms precise.

Definition 8–11. A *parallelogram* is a quadrilateral having two pairs of parallel sides. A *rectangle* is a parallelogram with a right angle. A *square* is a rectangle with two congruent perpendicular sides. A *rhombus* is an equilateral parallelogram.

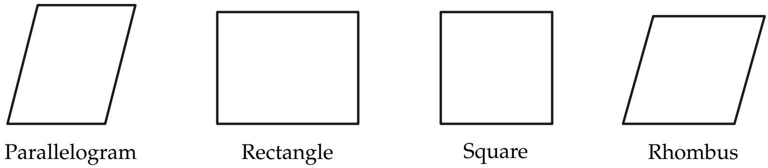

FIGURE 8–25

Our knowledge of parallels and these definitions make it possible to prove many results, as the following exercises will demonstrate.

Exercise Group 8–12

The exercises below present many interesting theorems concerning polygons. Give proofs for these theorems.

1. A diagonal of a parallelogram divides it into two congruent triangles.
2. Opposite sides of a parallelogram are congruent to each other.
3. The diagonals of a parallelogram bisect each other.
4. Opposite angles of a parallelogram are congruent.
5. Consecutive angles of a parallelogram are supplementary to each other.
6. Every angle of a rectangle is right.
7. All sides of a square are congruent.
8. The diagonals of a rectangle are congruent.

9. The diagonals of a square are perpendicular to each other.
10. The diagonals of a rhombus are perpendicular.
11. If the diagonals of a parallelogram are congruent, the parallelogram is a rectangle.
12. If the diagonals of a rectangle are perpendicular, the rectangle is a square.
13. The quadrilateral formed by connecting midpoints of the sides of a parallelogram is also a parallelogram.
14. The parallelogram formed by connecting midpoints of the sides of a square is also a square.
15. If the diagonals of a quadrilateral bisect each other, the quadrilateral is a parallelogram.
16. If a quadrilateral has a pair of sides congruent and parallel to each other, it is a parallelogram.
17. A parallelogram has one angle of $70°$. Give the number of degrees in each angle.
18. One angle of a parallelogram is twice as large as a second angle. Determine the number of degrees in each angle of the parallelogram.
19. A diagonal of a rhombus bisects two angles of the rhombus.
20. If a rectangle is not a square, a diagonal does not bisect an angle of the rectangle.
21. If each angle of a quadrilateral is congruent to the angle opposite it, then the quadrilateral is a parallelogram.
22. Is a rhombus a regular polygon?
23. For every quadrilateral, it is true that either the bisectors of two consecutive angles of the quadrilateral are perpendicular, or the quadrilateral is not a parallelogram.

24. If a diagonal of a parallelogram bisects one angle of the parallelogram, it bisects a second angle, and the parallelogram is a rhombus.

25. If a parallelogram is not a rhombus, then the four bisectors of the angles of the parallelogram are not concurrent.

26. If a parallelogram is not a rhombus, then the four bisectors of the angles intersect to form a rectangle.

27. The four bisectors of the angles of a rectangle that is not a square intersect to form a square.

28. How many degrees are there in the angle formed by the two diagonals drawn from one vertex of a regular pentagon?

29. In Fig. 8–26, $DE \parallel AC$, and $DF \parallel AB$. Use the figure to prove that the sum of the angles of $\triangle ABC$ is a straight angle.

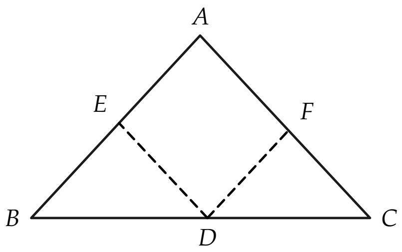

FIGURE 8–26

30. In the regular hexagon $ABCDEF$, prove that diagonal FD is

perpendicular to side DC (Fig. 8–27).

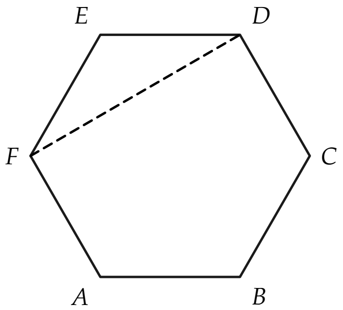

FIGURE 8–27

31. Prove that $\overline{FC} = 2 \cdot \overline{DC}$ in Fig. 8–27. Show that the diagonals FC and AD bisect each other.

REVIEW OF CHAPTER 8

This chapter has dealt with consequences of the parallel postulate, by far the most famous of all the postulates of Euclid. (Euclid did not state it exactly as we have given it.) It was through the careful investigation of this postulate that non-Euclidean geometries were invented early in the 19th century. This investigation also led indirectly to the first set of correct postulates from which the theorems of Euclidean geometry could be rigorously proved. These postulates, presented about 1900 by the German mathematician David Hilbert, are equivalent to the ones that we have listed in this text. We list below some of the most important consequences of the parallel postulate shown in this chapter.

> If two lines are parallel to one line, they are parallel to each other.

When parallel lines are cut by a line, alternate interior angles are congruent.

The sum of the angles of any triangle is a straight angle.

The sum of the interior angles of any convex polygon is $(n-2)$ straight angles.

Several new terms were defined in this chapter. In addition, precise definitions of some familiar concepts were formulated. (For example, though we have referred freely to squares and rectangles in earlier chapters, they had not yet been exactly defined. Indeed, it is not even possible to prove that squares and rectangles exist until the parallel postulate is available.) You should know the definitions given in this chapter for the following:

alternate interior angles	octagon
corresponding angles	n-gon
polygon	regular polygon
simple polygon	parallelogram
convex polygon	rectangle
parallel lines	square
pentagon	rhombus
hexagon	diagonal
equilateral polygon	equiangular polygon

ALGEBRA REVIEW

1. Two angles of a triangle are congruent, and the measure of each exceeds that of the third angle by $10°$. Determine the number of degrees in each angle of the triangle.
2. In $\triangle ABC$, the sum of $\angle B$ and an exterior angle at A has a measure of $170°$. The sum of this exterior angle and $\angle C$ is $160°$. Determine the measure of each angle of the triangle.

3. An exterior angle of a triangle has a measure of 117°. One angle of this triangle has a measure of 42°. Determine the degree measure of each angle of the triangle.

4. Prove that there is no $\triangle ABC$ such that the measure of an exterior angle at A is 103° and an exterior angle at B is 74°.

5. Diagonal AC of parallelogram $ABCD$ forms with sides AB and AD two angles whose measures differ by 10°. The measure of $\angle B$ is 110°. Determine the measure of $\angle A$.

6. Two angles of a triangle are complementary. Prove that the measure of the third angle is 90°.

7. The measure of the vertex angle of an isosceles triangle is 84°. Find the measure of each angle of the triangle.

8. The first angle of a triangle has a measure three-fourths of the second. The first angle is also one-third of the sum of the other two. Compute each angle.

9. In Fig. 8–28 the letters represent the degree measures of the indicated angles. Prove that $x + y + t = r + s + z$.

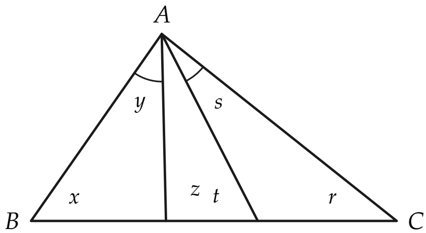

FIGURE 8–28

10. In Fig. 8–29, $AB \cong BC$, $BE \cong BD$, and the measure of $\angle CBD$

is 40°. Determine the measure of $\angle ADE$.

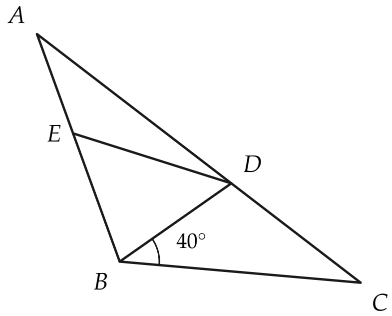

FIGURE 8–29

REVIEW EXERCISES, CHAPTERS 1 THROUGH 8

1. Give truth values of α, β, and γ for which a statement of the form (α or β) and γ is true; is false.

2. Can you find truth values of α, β, and γ for which (α or β) and γ is false, and α or (β and γ) is true? Can you find truth values for which (α or β) and γ is true, and α or (β and γ) is false? Is (α or β) and $\gamma \to \alpha$ or (β and γ) true? Is α or (β and γ) $\to$ (α or β) and γ true?

3. Formulate an "if…, then…" statement equivalent to (a) no fishermen lie, (b) only fishermen lie, (c) fishermen necessarily lie.

4. Give the contrapositive of each of your implications in Exercise 3.

5. One set of points contains, among others, the points A, B, C, and D. A second set of points contains A, B, and D, but does

not contain the point C. Prove that these two sets of points cannot both be lines.

6. Point C is in the interior of $\angle AOB$. Is every point of ray OC in the interior of this angle?

7. Define *interior of a triangle* in at least two ways.

8. Explain what is meant by saying that *congruence of segments* has the *reflexive* property; the *symmetric* property; the *transitive* property.

9. Explain why one of these statements is true, and the other false:
(a) $AB \cong CD \to AB = CD$, (b) $AB = CD \to AB \cong CD$.

10. Figure 8–30 indicates that $AB \cong A'B'$, and ABC. Using only the definitions of *greater than* and *less than*, does the figure indicate that $AC > A'B'$, or that $A'B' < AC$?

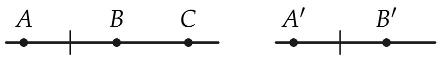

FIGURE 8–30

11. The length of AB, measured with CD as unit, is 17.23. What is Archimedes' integer?

12. The length of CD in Exercise 11, as measured by EF, is 10. If AB is measured by EF, what is its length? What is Archimedes' integer in this case?

13. Express the rational number $\frac{3}{19}$ as a repeating decimal. Find the rational number whose decimal representation is the repeating decimal $0.272727\ldots$; $0.027027027\ldots$.

14. State the S.A.S. postulate for triangles.

15. Two lines that intersect to form congruent adjacent angles are said to be perpendicular to each other. Does this definition in

itself prove that there is a pair of perpendicular lines in the plane? How do we know that right angles exist?

16. Are there points A, B, and C, such that $\overline{AB} = 5$, $\overline{BC} = 7$, and $\overline{AC} = 13$?

17. What can be said about the position of the points R, S, and T, if $\overline{RS} = 8$, $\overline{ST} = 19$, and $\overline{RT} = 11$?

18. Consider a circle whose center is A and which contains the point B.

    (a) If $\overline{AX} > \overline{AB}$, what can be said about point X?

    (b) If P and Q are on the circle, how do we refer to $\angle PAQ$? to segment PQ?

    (c) If O and C are on the circle, how do we refer to $\angle AOC$?

    (d) If R and S are points of the circle, and A is on the segment RS, what do we call segment RS?

19. What does it mean to say that two rays have the same direction?

*20. Without using the parallel postulate or any of its consequences, prove the following theorem.

    If in triangles ABC and $A'B'C'$, $AB \cong A'B'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$, then $\triangle ABC \cong \triangle A'B'C'$.

*21. Without using the parallel postulate, prove the following theorem, which relates to Fig. 8–31.

    If $\angle CAB$ and $\angle ABD$ are right, $CA \cong DB$, and M and N are midpoints of CD and AB, respectively, then MN is perpendicular both to AB and to CD.

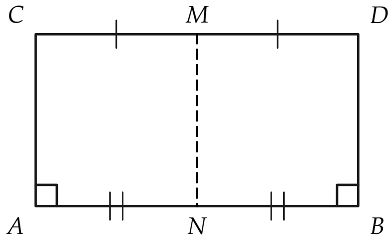

FIGURE 8–31

*22. Carry through the proofs of Exercises 20 and 21 using the parallel postulate.

*23. Euclid's formulation of the parallel postulate is different from ours. The following statement is essentially the postulate that Euclid used:

> *If the points X and Y are on one side of line AB, and if the sum of $\angle XAB$ and $\angle YBA$ is less than a straight angle, then the rays AX and BY intersect.*

Figure 8–32 illustrates Euclid's parallel postulate. (a) Prove Euclid's parallel postulate as a theorem, using our parallel postulate. (b) Assuming the truth of Euclid's parallel postulate, prove ours as a theorem.

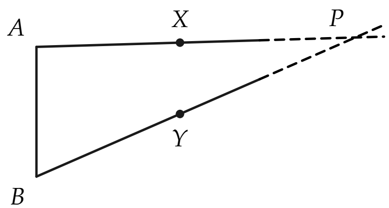

FIGURE 8–32